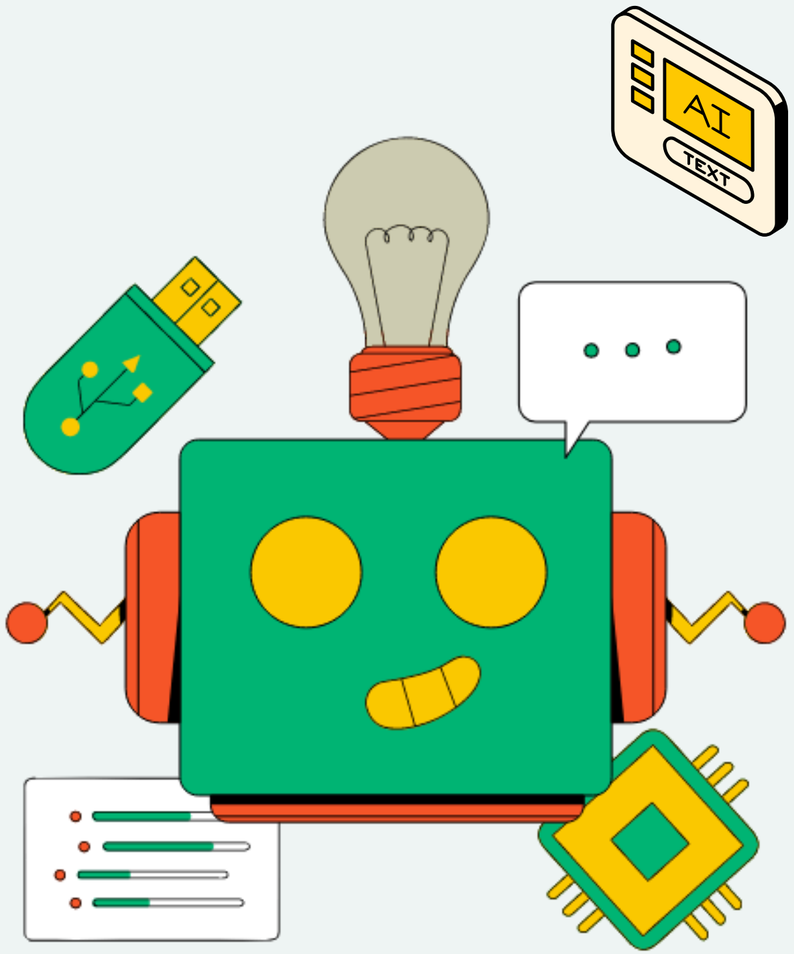
H250033



A Retrieval-Augmented Generation Chatbot with Fine-Tuned Large Language Models for Context-Aware Mental and Physical Health Solutions

Shu-Han Chuang, Miao-Yin Chen

Department of Data Science, Soochow University

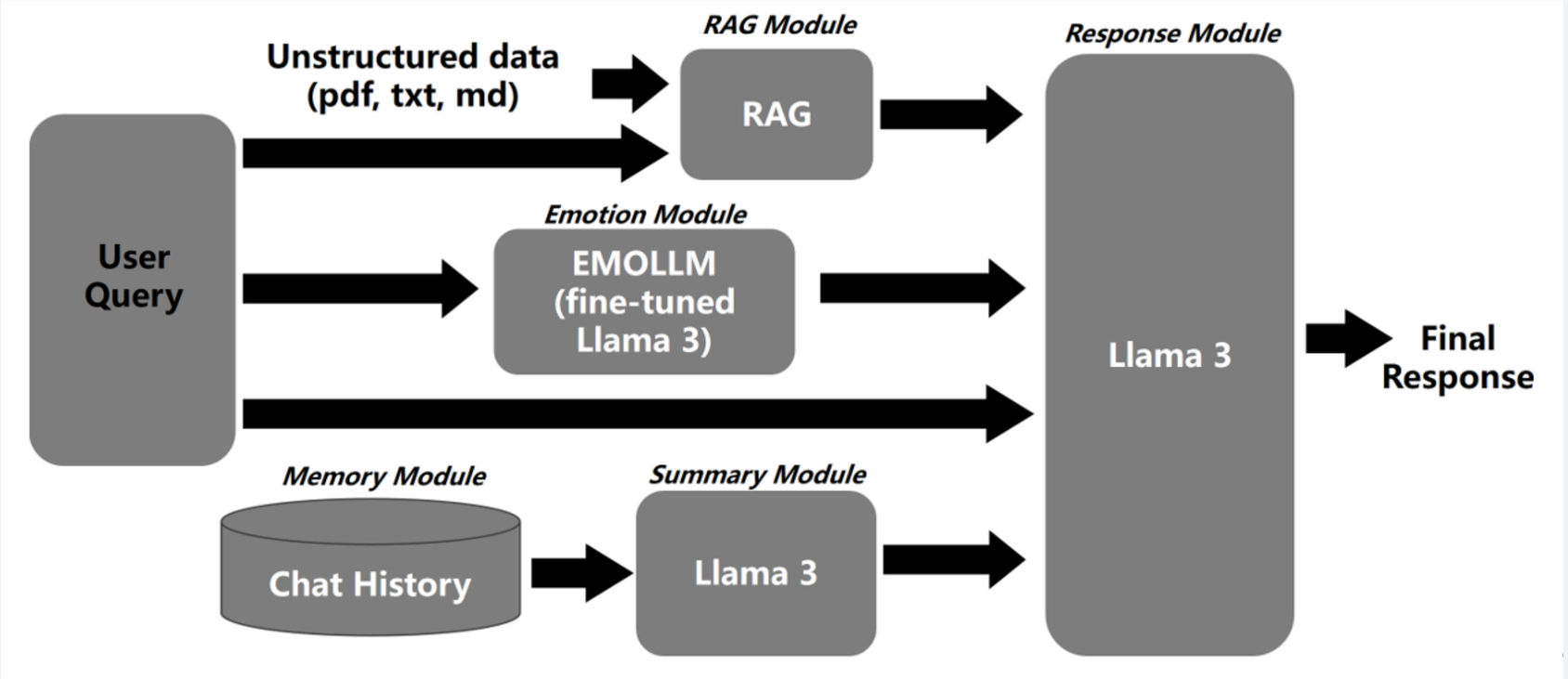
INTRODUCTION

- Traditional healthcare communication is inefficient and frustrating.
- Chatbots enhance accessibility and reduce administrative burden.
- Chatbot types:
  - Rule-based: fixed but inflexible.
  - Retrieval-based: context-aware but limited.
  - Generative: creative, but hallucinate and lack coherence.
- LLMs face domain-specific challenges (e.g. hallucination).
- RAG has real-time factual retrieval and generation.
- Empathy is also critical, especially in mental health.
- We propose a hybrid chatbot combining factual accuracy and emotional intelligence.

II. METHOD

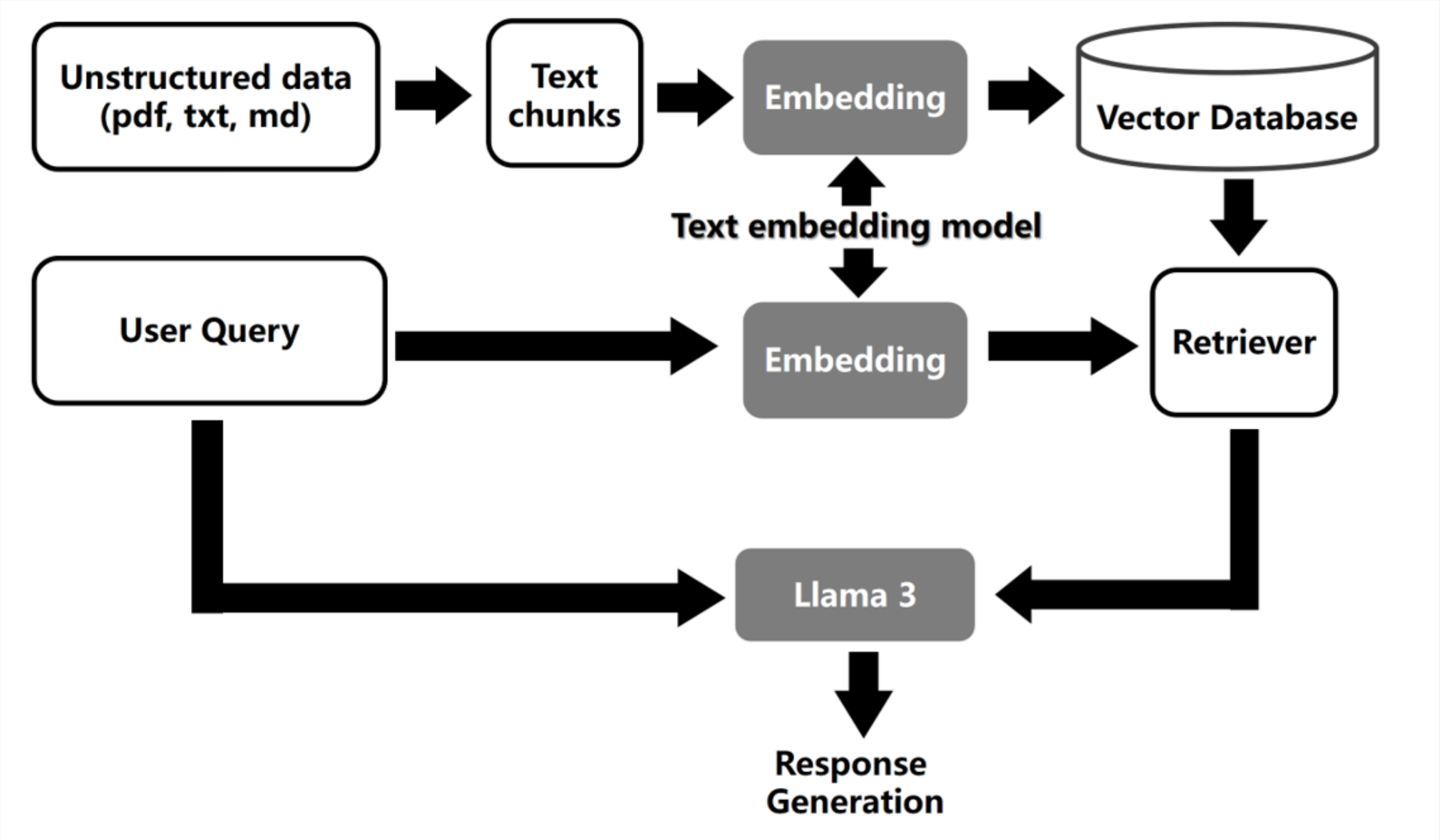
SYTSEM ARCHITECTRUE AND WORKFLOW

- Based on open-source LLaMA 3 with hybrid components:
  - RAG Module: Retrieves medical content from PDFs, web, JSON, markdown, etc.
  - Emotion Module: Detects and responds with empathy, trained on mental health dialogue.
  - Memory Module: Stores and recalls user history across sessions for personalized interactions.



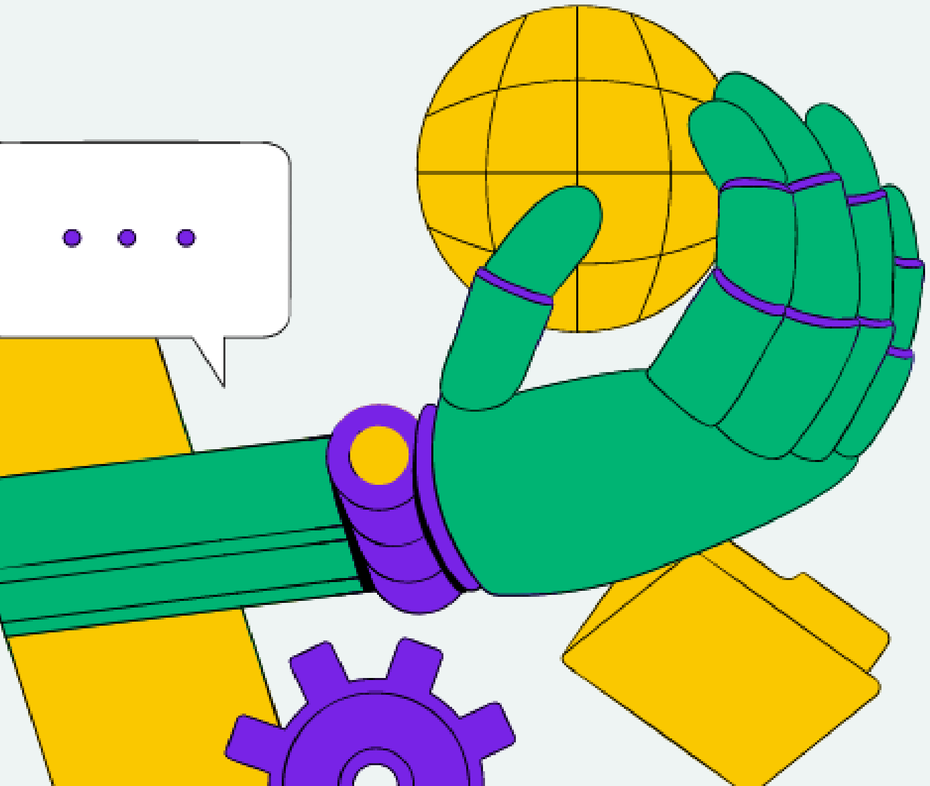
RAG FOR PHYSICAL HEALTHCARE DOMAIN

- Main steps:
  - The document corpus is segmented into overlapping text chunks.
  - Retrieval: Semantic search via MiniLM Embedding, and save to FAISS vector database with its metadata (title, source, timestamp).
  - Generation: Synthesizes LLM responses based on retrieved content.
- User query and vector database match:
  - Uses Maximum Marginal Relevance (MMR) to reduce redundancy and enhance diversity.



- Information Retrieval with RAG Workflow

Emotion-Aware LLM (EmoLLM)



- Trained on emotionally annotated multi-turn mental health dialogues.
- Apply LoRA for efficient fine-tuning on LLaMA 3.
- Responds with empathy by learning affective patterns.
- EmoLLM + RAG work together to deliver both factual and emotionally responses.

CONTEXTUAL MEMORY MODULE

- Maintains dialogue history using Google Firestore.
- We summarize past interactions to meet token limits using LLM.
- Integrates summaries into prompt.

"You are a friendly and logical assistant. Based on the following {summarized conversation history} and {question}, as well as the two answers provided by {RAG} and {EmoLLM}, please merge them into one natural, emotionally expressive response in Traditional Chinese within 200 characters."

PERFORMANCE OF OUR HYBRID SYSTEM

- Evaluated via LINE chatbot with about 100 real user queries over 2 weeks
- Each response was rated on a 5-point Likert scale across six dimensions:

Perceived Helpfulness

Measures whether the system's advice provides informative, actionable, and contextually useful.

Naturalness and Clarity

Assesses fluency, grammatical coherence, and overall ease of understanding.

Context Awareness

Evaluates the system’s ability to preserve from conversational turns.

PERFORMANCE OF OUR HYBRID SYSTEM

- Evaluated via LINE chatbot with about 100 real user queries over 2 weeks
- Each response was rated on a 5-point Likert scale across six dimensions:

Trust and Credibility

Represents user perceptions of the response's reliability.

Emotional Resonance

Measures the chatbot's sensitivity to emotional cues and its capacity to deliver empathetic, human-like dialogue.

Satisfaction and Retention Intention

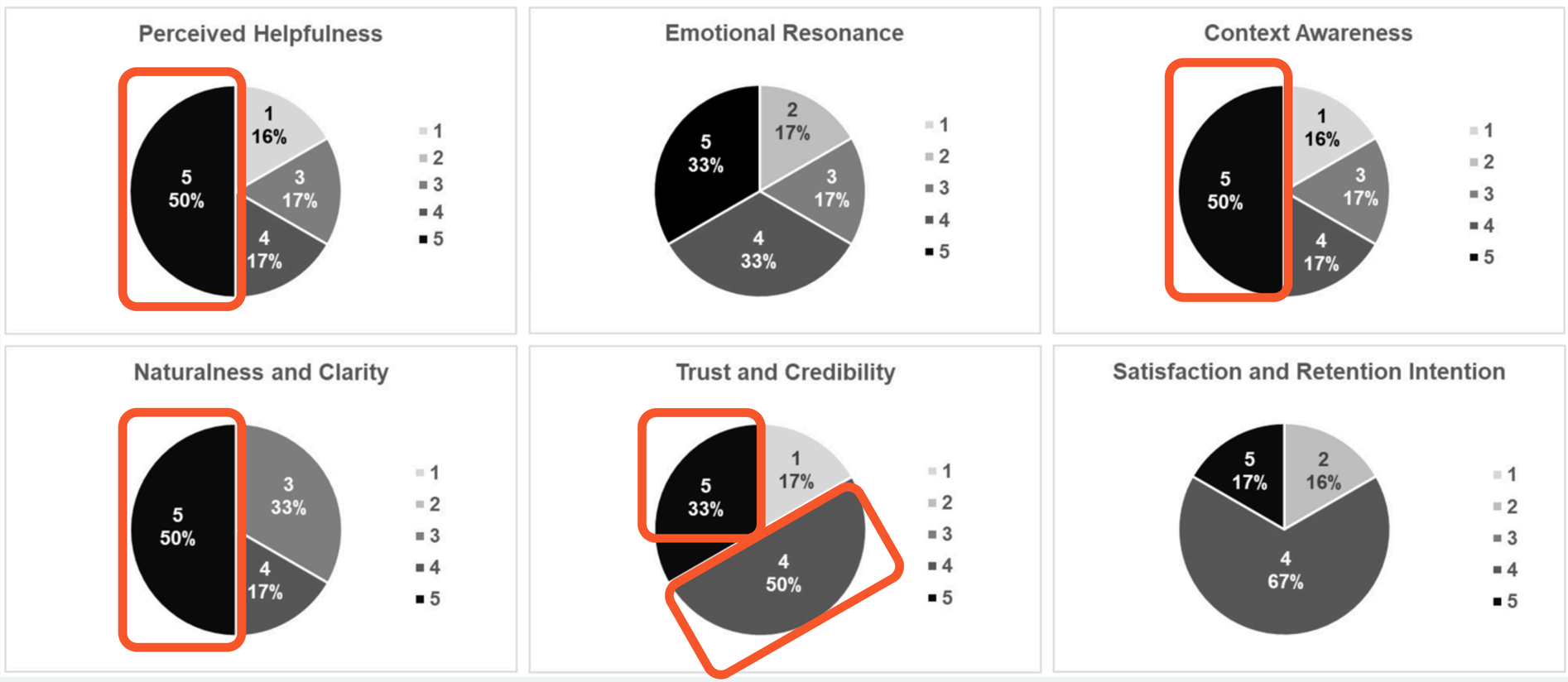
Reflects the user’s overall satisfaction to engage with the system.

III. EXPERIMENTS AND RESULTS

DATA SOURCES

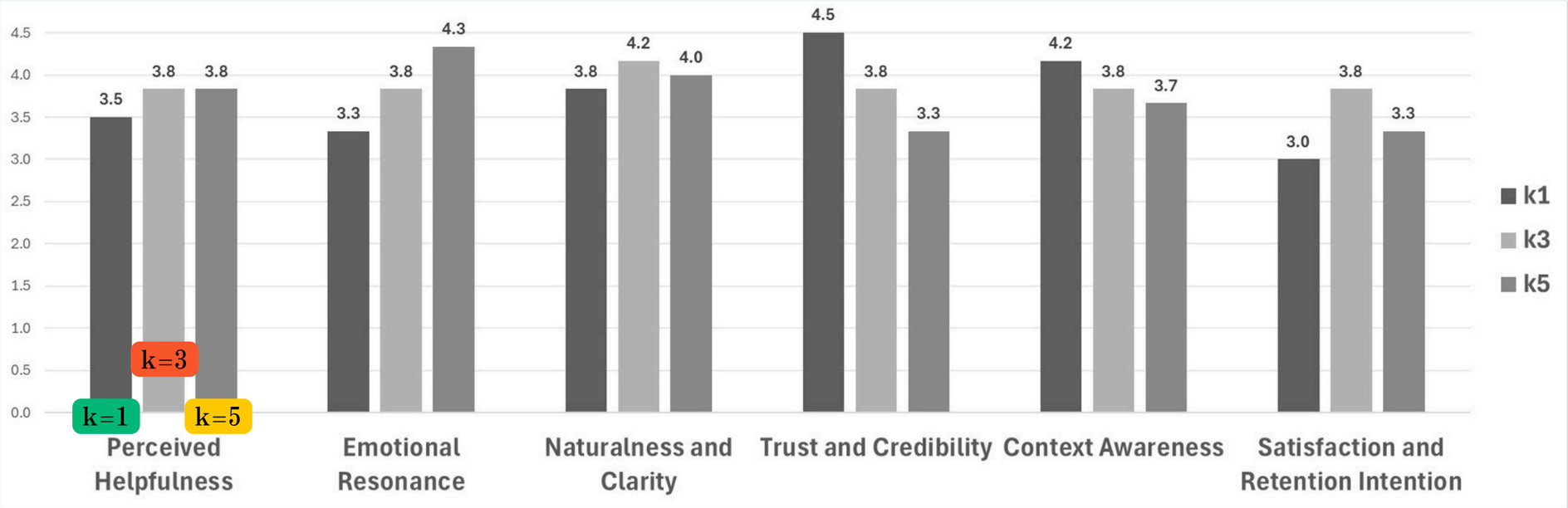
- Collected from:
  - Tri-Service General Hospital
  - Taiwan CDC & Health Promotion Administration
- 1,215 curated Chinese-language health documents (physical + mental health).
- Preprocessed into 1024-token chunks with 200-token overlap for semantic continuity.

- The evaluation results from user testing:
  - High ratings in Helpfulness, Context Awareness, Clarity, Trust
  - Moderate performance in Emotional Resonance
  - Validated via inter-rater reliability (Cohen's Kappa = 0.65)



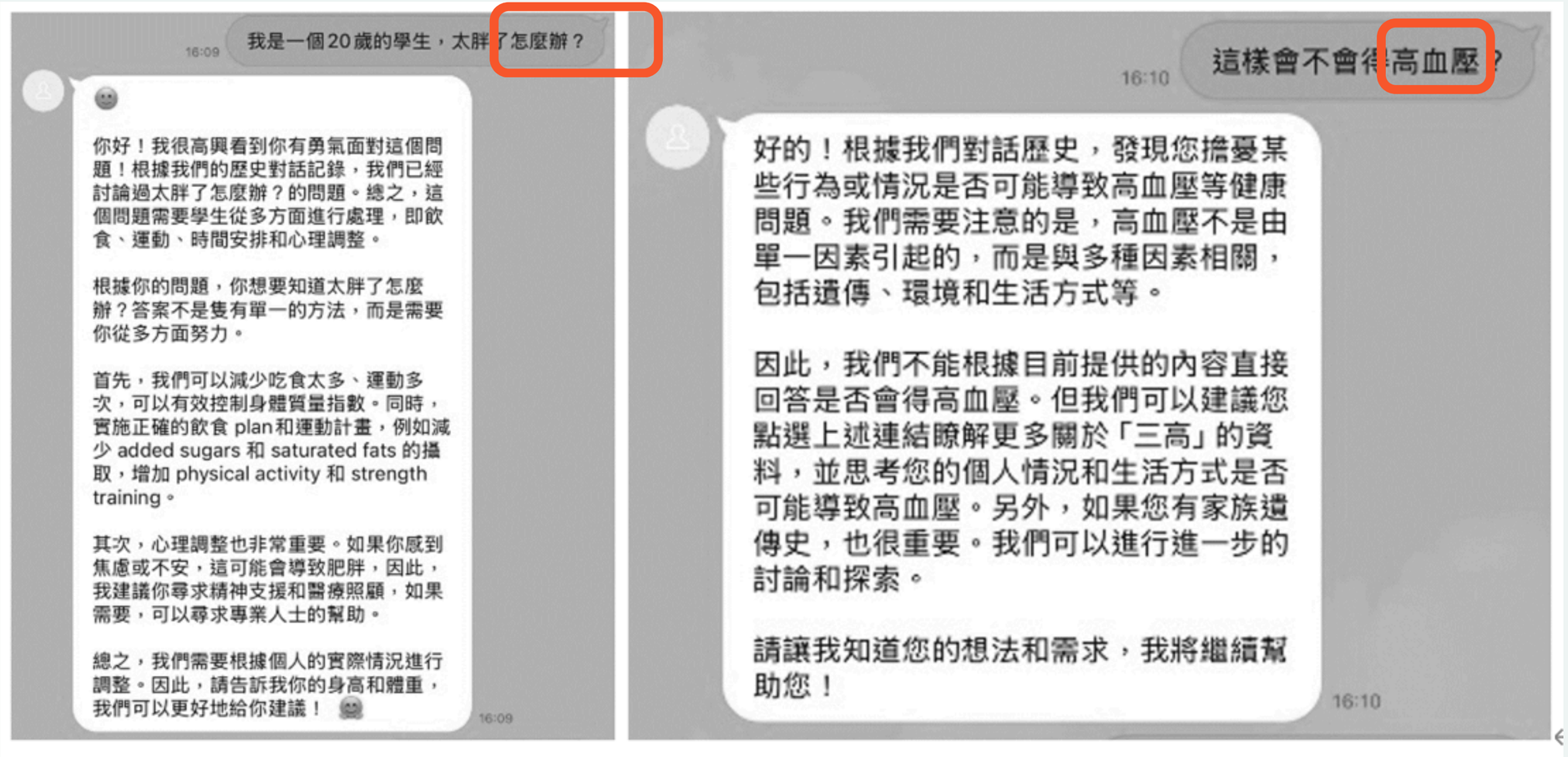
TOP-K RETRIEVAL EVALUATION

- Tested k = 1, 3, 5 in RAG module:
  - k = 3: Best overall performance across metrics (balanced input)
  - k = 1: Highest Trust but weakest in Emotional Resonance
  - k = 5: Strong Emotional Resonance but reduced clarity & trust

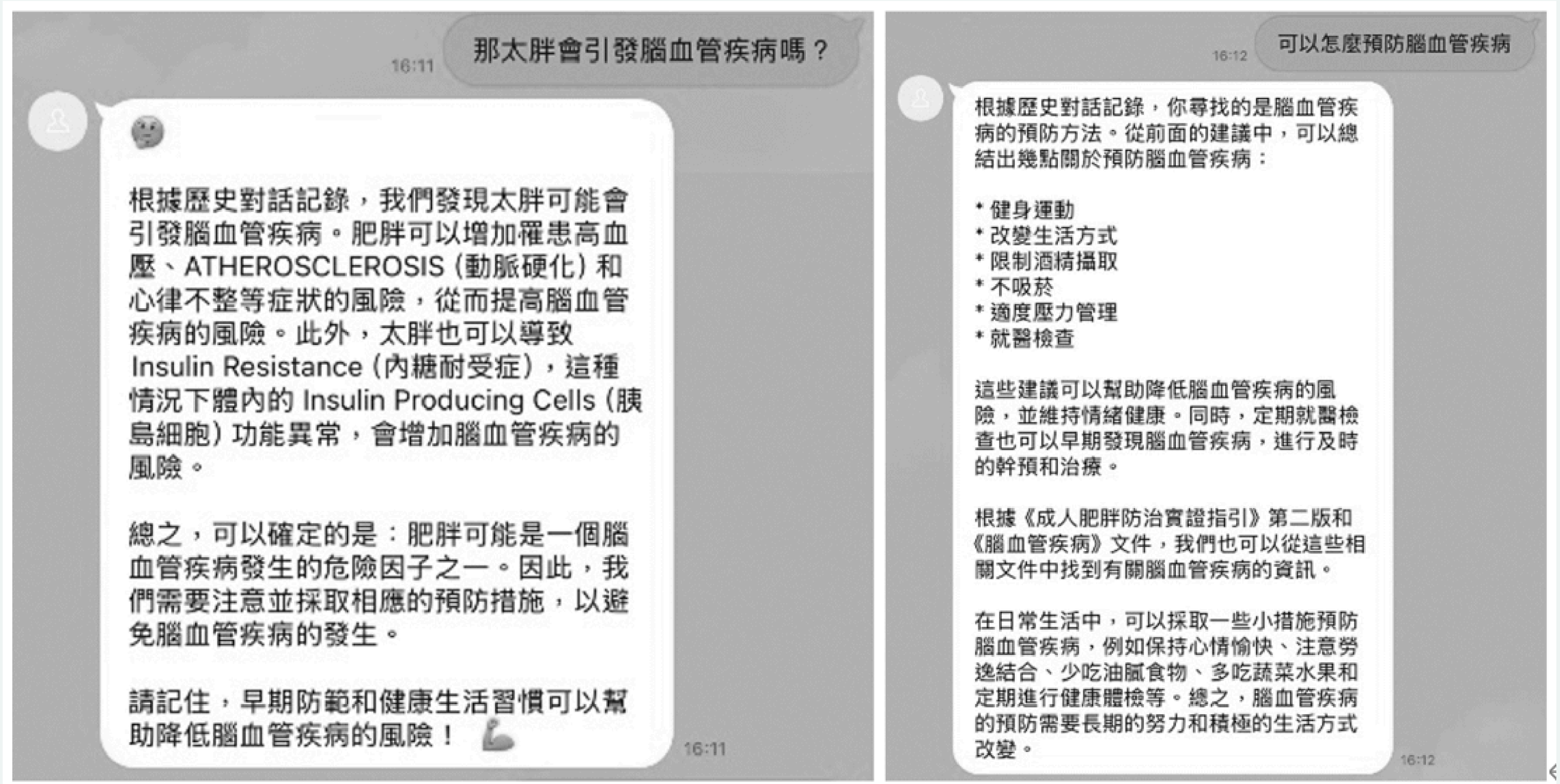


- Conclusion: k = 3 offers better trade-off between relevance and emotional context.

- An example of a context-aware chatbot conversation addressing physical health concerns

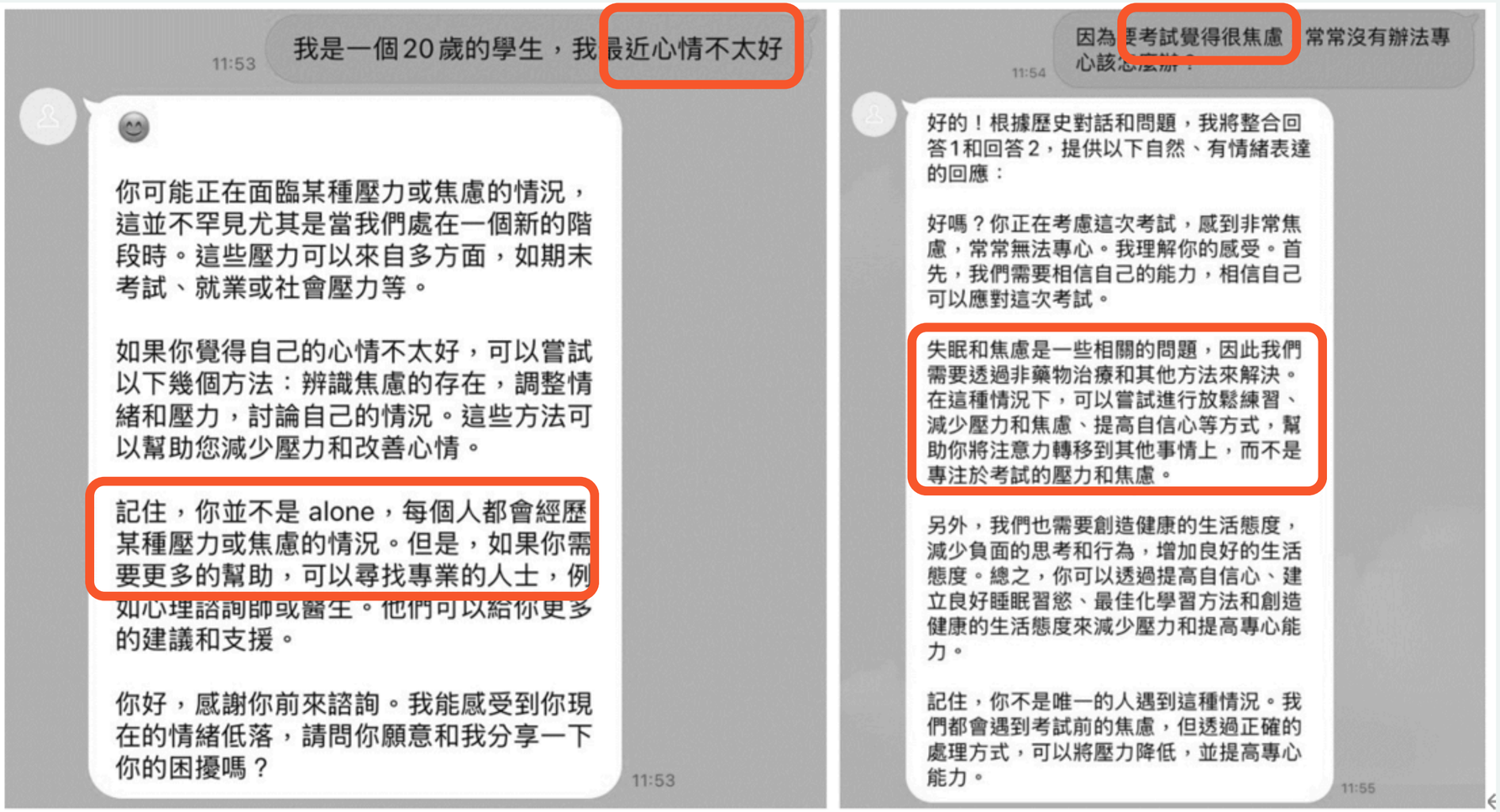


Its about fat and high blood pressure associations, the chatbot emphasizes the importance of lifestyle management. By delivering evidence-based advice, the system educates and empowers users to take an active role in managing their health.



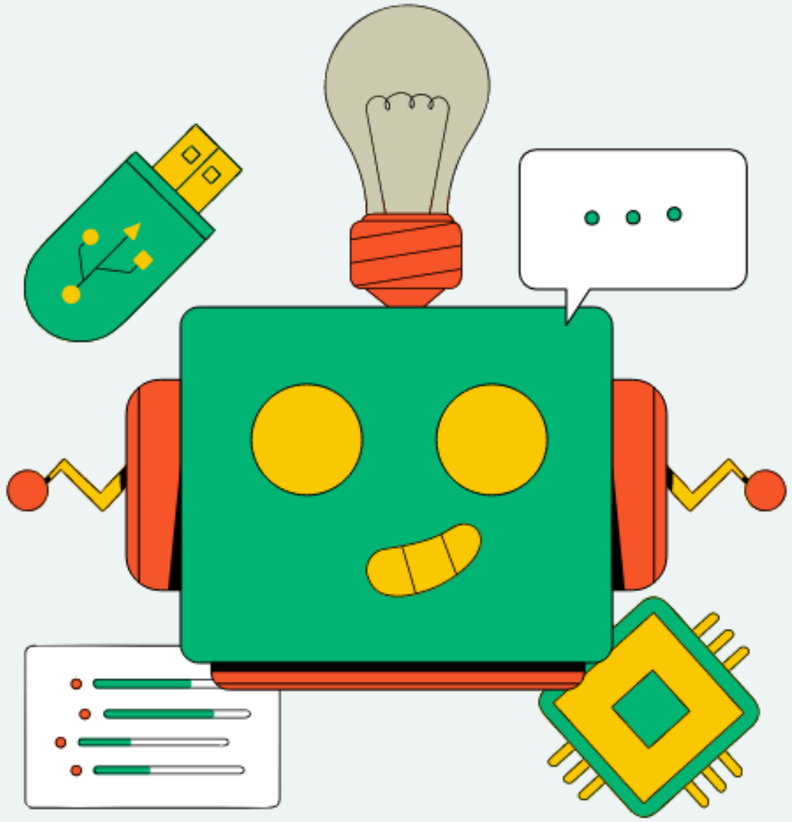
- An example of a context-aware chatbot conversation addressing physical health concerns with memory integration

- An example of a context-aware chatbot conversation addressing mental health concerns



The examples express emotions such as sadness due to academic stress. The LINE chatbot responds empathetically, validating users' feelings and offering practical strategies, including relaxation techniques, and emotional regulation.

CONCLUSION

- We developed a hybrid LLM healthcare chatbot integrating:
  - RAG: Real-time, reliable information
  - EmoLLM: Emotion-aware communication
  - Memory module: Personalized, multi-turn continuity
- We validated in real-world settings via LINE chatbot interface
- We demonstrate high user satisfaction and potential for:
  - Physical and mental health support
  - Emotionally intelligent healthcare systems

THANKS FOR YOUR ATTENTION.